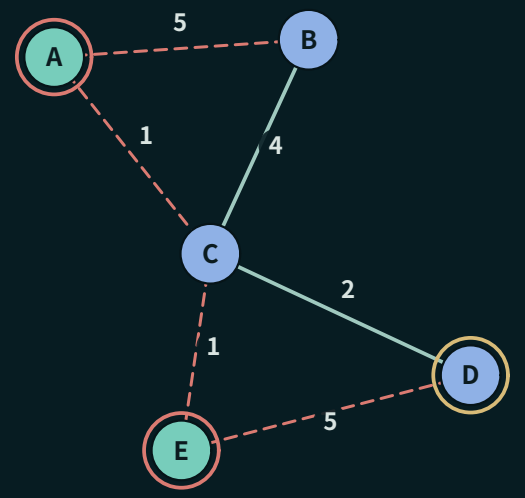
The legal move

Fiduccia–Mattheyses refinement · single moves and the balance window

Learning Clustering · original author stephQ-lab (Issue #7 attachment)

The problem, and how it differs from KL

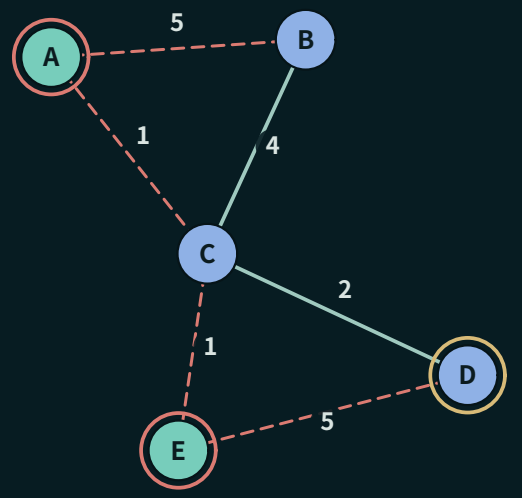
- Input: a graph (or a multi-pin netlist), an initial bisection, maxDiff
- Move: one node at a time
- Legal: after the move both sides are nonempty and $|n_0 - n_1| \leqslant$ maxDiff
- Output: the trial moves, cumulative gains, the best prefix, the accepted bisection
- Review: gain, lock, trial and best prefix are as in Module 07



Node	Side	Gain	Legal
A	0	+6	no
B	1	+1	yes
C	1	-4	yes
D	1	+3	yes
E	0	+6	no

The largest may not be allowed to move

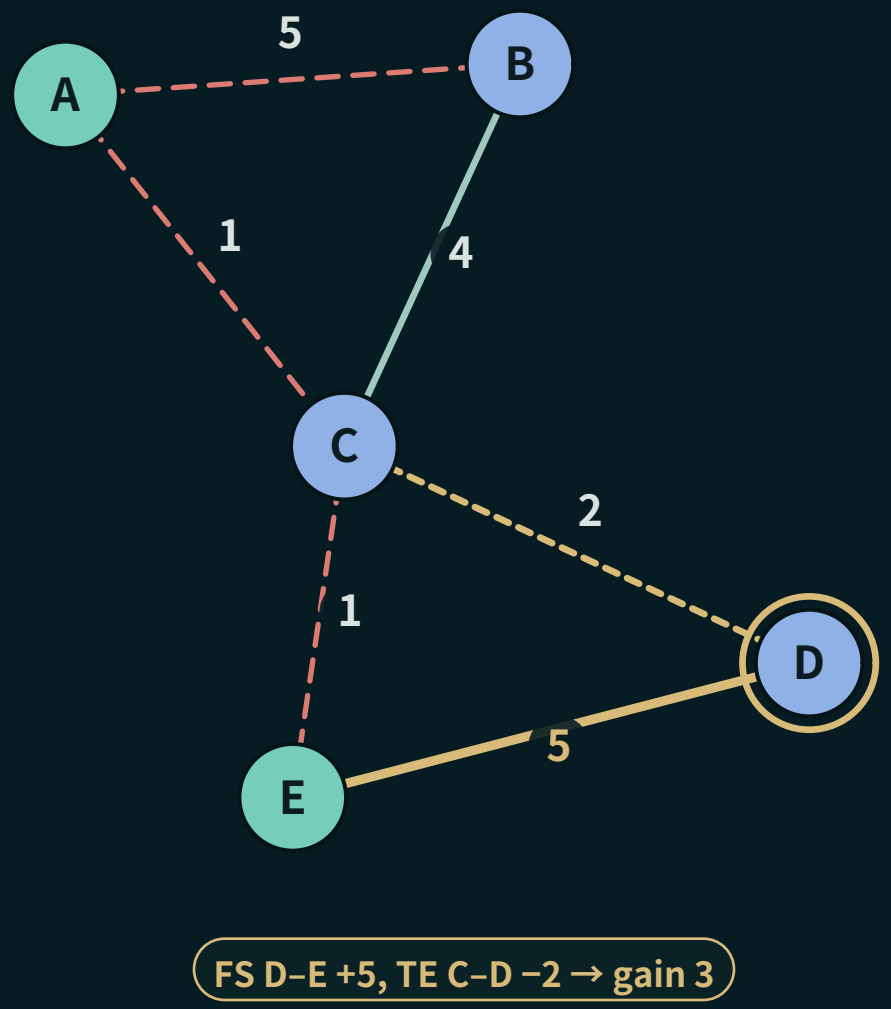
- Seed AE | BCD, maxDiff = 1: only 2:3 or 3:2
- A and E gain +6, but sit on the left, which has 2 nodes
- Moving either leaves 1 : 4: not legal
- Legal candidates B +1, C −4, D +3 → choose D



Node	Side	Gain	Legal
A	0	+6	no
B	1	+1	yes
C	1	-4	yes
D	1	+3	yes
E	0	+6	no

Splitting a gain: FS and TE

- FS: v is a net's only pin on its side → +w
- TE: a net has no pin on the target side → −w
- D: FS = D–E (5), TE = C–D (2), gain 3
- On 2-pin edges: FS − TE = external − internal



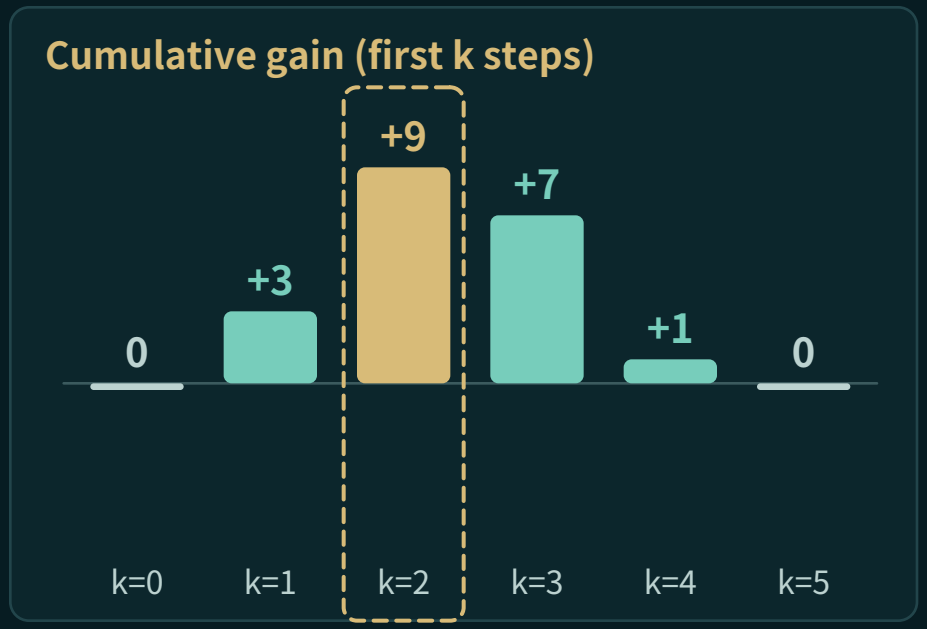
One full pass

- D +3 (cut 9) → the left has 3 nodes, and A becomes legal
- A +6 (cut 3)
- Then C −2, E −6, B −1: tried anyway
- The cut runs 12, 9, 3, 5, 11, 12

Step	Move	Gain	cut	Cum.
1	D	+3	9	+3
2	A	+6	3	+9
3	C	-2	5	+7
4	E	-6	11	+1
5	B	-1	12	0

The best prefix and roll-back

- Cumulative gains 3, 9, 7, 1, 0
- k* = 2: apply only D and A
- Accept ABC | DE, cut 3
- The trials' end at cut 12 is discarded; pass 2's prefix is empty, so stop



Accept the first 2: ABC | DE, cut 3

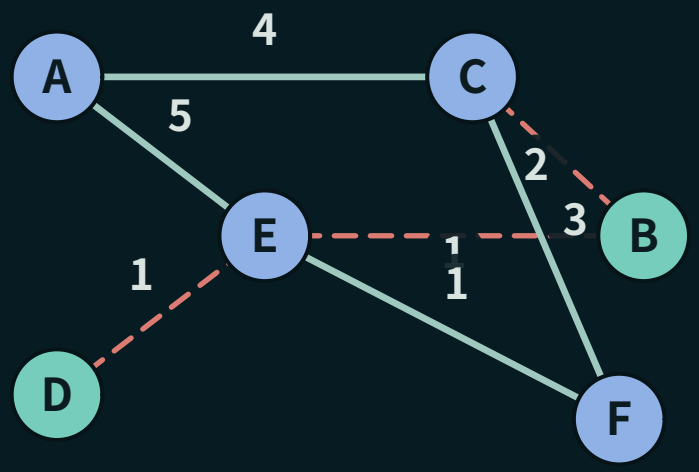
Auditing the attachment's hillclimb claim

- The text: initial cut 9, legal gains all $\leqslant 0$, final 4
- From the attachment's data: initial cut 18
- E is legal, with gain +10
- Pass 1's best 8 = the exhaustive balanced optimum 8
- Text and data disagree: a regression test only, not teaching material

	The text	From its data
Initial cut	9	18
Legal gains	all $\leqslant 0$	E legal, +10
Final cut	4	pass 1: 8
Exhaustive optimum		8

The replacement: negative first, then positive

- 6 nodes, seed ADE | BCF, maxDiff = 2, cut 6
- At the start every legal gain ⩽ 0: hill climbing stops at 6
- FM: A −1, B −1 (cut 8), then E +4 (cut 4)
- Keep 3 steps: cut 4 = the exhaustive balanced optimum
- Found by a seeded search, verified exhaustively in the tests

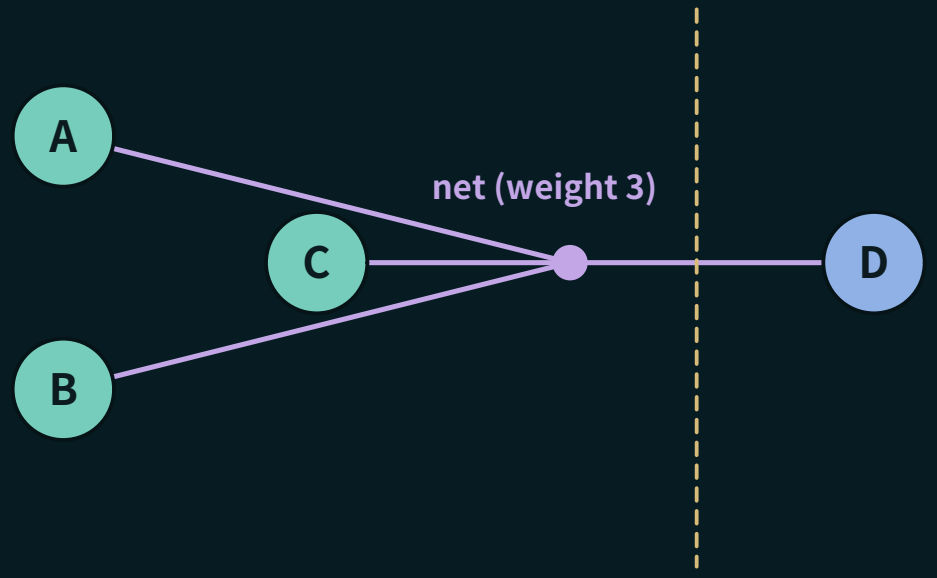


Step	Move	Gain	cut
1	A	-1	7
2	B	-1	8
3	E	+4	4

Keep 3 steps: cut 6 → 4 = the exhaustive optimum

Buckets, complexity and multi-pin nets

- The paper: bucket lists + a max-gain pointer → O(P) per pass
- This course: gains update incrementally, but selection still scans: not O(P)
- Multi-pin nets: decide FS/TE by counting pins per side
- net{A,B,C,D}, D on the right: A gains 0, D +3; a clique would give −1
- The attachment's "hypergraph" case has only 2-pin edges



Counting pins: A 0, D +3; as a clique: A −1

The pseudocode, line by line

- Line 2: compute gains by counting per net (gain)
- Lines 3–4: take the largest among legal candidates only (legal_moves)
- Lines 5–6: trial, lock, incremental update
- Lines 7–8: best prefix and roll-back; line 9: stop on the empty prefix

```
1  repeat: unlock all, remember the start
2    count pins per net, compute every gain
3    while a legal unlocked node exists:
4      v ← the legal candidate with the largest gain
5      move v (a trial), lock v
6      update only the affected nodes' gains
7    k* ← the best prefix (0 allowed)
8    back to the start, apply only the first k*
9  until k* = 0
```

Common mistakes

- Picking the largest gain outright (ignoring balance)
- Copying the attachment's hillclimb claim
- Taking "buckets drawn" for "linear time implemented"
- Calling a 2-pin example a hypergraph
- Thinking count balance is area balance

Summary

Filter for legality, then take the largest; try negative gains too; accept only the best prefix.

Verify examples by independent computation; buckets on screen are not a linear-time implementation.

Next module: the goal changes from the cut to delay, and nodes may be copied.